

SYBSC semster-IV

Python Programming-II (PR)

[How to start working with Python IDE](#)

Follow the instructions below to start a jupyter notebook in the Linux environment.

Step 1:

Run following command on Linux terminal:

1. `source pyenv/bin/activate`
2. `jupyter lab`

Step 2: Start new Python 3 notebook from Jupyter and Rename notebook as per exercise

NOTE: install numpy library using following statement in script (if required)

pip install numpy

ASSIGNMENT-1(A)

Aim: basic use of Numpy library in PYTHON

Examples:

1.
`import numpy as np`

`# create a list named list1`
`list1 = [2, 4, 6, 8]`

`# create numpy array using list1`
`array1 = np.array(list1)`

`print(array1)`
2.
`import numpy as np`

`# create numpy array using a list`
`array1 = np.array([2, 4, 6, 8])`
`print(array1)`
- 3.

```
import numpy as np
```

```
# create an array with 4 elements filled with zeros  
array1 = np.zeros(4)
```

```
print(array1)
```

4.

```
import numpy as np
```

```
# create an array with values from 0 to 4  
array1 = np.arange(5)
```

```
print("Using np.arange(5):", array1)
```

```
# create an array with values from 1 to 8 with a step of 2  
array2 = np.arange(1, 9, 2)
```

```
print("Using np.arange(1, 9, 2):", array2)
```

5.

```
import numpy as np
```

```
# generate an array of 5 random numbers  
array1 = np.random.rand(5)
```

```
print(array1)
```

6.

```
import numpy as np
```

```
# create an empty array of length 4  
array1 = np.empty(4)
```

```
print(array1)
```

7.

```
import numpy as np
```

```
# create a 2D array with 2 rows and 4 columns  
array1 = np.array([[1, 2, 3, 4],  
                  [5, 6, 7, 8]])
```

```
print(array1)
```

8.

```
import numpy as np
```

```
# create a 3D array with 2 "slices", each of 3 rows and 4 columns  
array1 = np.array([[[1, 2, 3, 4],  
                   [5, 6, 7, 8],  
                   [9, 10, 11, 12]],  
                  [[13, 14, 15, 16],  
                   [17, 18, 19, 20],  
                   [21, 22, 23, 24]]])
```

```

print(array1)

9.
import numpy as np

# create 2D array with 2 rows and 3 columns filled with zeros
array1 = np.zeros((2, 3))

print("2-D Array: ")
print(array1)

# create 3D array with dimensions 2x3x4 filled with zeros
array2 = np.zeros((2, 3, 4))

print("\n3-D Array: ")
print(array2)

10.
import numpy as np

# Create a 2-D array with elements initialized to 5
numpy_array = np.full((2, 2), 5)

print("Array:", numpy_array)

11.
import numpy as np

# create an array of 32-bit integers
array1 = np.array([1, 3, 7], dtype='int32')

print(array1, array1.dtype)

```

Basic:

1. Write a NumPy program to create an array of integers from 30 to 70. Display array.
2. Write a NumPy program to create an array of all even integers from 30 to 70. Display array and size.
3. Write a NumPy program to create an array of integers from 21 to 30 with 2-D
4. Write a NumPy program to create an array of floating numbers with 10 different elements.
5. Write a numpy program to create 3-D with different numbers.

MODERATE:

6. Create at least 2 matrices of 2x 2 with different integer values. And find sum of them. Display result also.
7. Create matrix of 3x5 with different integer values. Display array elements and convert it into 5x3. display again.
8. Create 2-D array with integer values and do access of each an element.
9. Create 1 matrix with all elements have same value as 1 and 2nd matrix with all elements have same value as 5. display both matrix.
10. Create a matrix 4x4 with all values are zero.

ASSIGNMENT-1(B)

AIM: use numpy library for array operations

1. Create a 1D NumPy array of numbers from 0 to 9
2. Write a code to convert a 1D array to a 2D array with 2 rows.
3. create an array of 7 different elements. and Print the following attributes of the array:
 - The shape of the array.
 - The number of array dimensions.
 - The size of each element in bytes.
4. Create a 1D array filled with zeros and another filled with ones.

5. Create a 1D array of 10 evenly spaced values between 5 and 50. [hint: use of `linspace()`]

output: `[ 5. 10. 15. 20. 25. 30. 35. 40. 45. 50.]`

6. take an array and Reverse a 1D NumPy array
7. Create a 3×3 identity matrix. [hint: use of `eye()`]
8. Create a 4×4 array and extract its first row and last column
9. create 3x3 matrix with sequence elements. Extract an array of odd rows and even columns from below NumPy array.
10. Write a code to Create a 4×4 NumPy array and extract the first two rows and first two columns.

Array:

```
[[ 1  2  3  4]
```

```
[ 5  6  7  8]
```

```
[ 9 10 11 12]
```

```
[13 14 15 16]]
```

First 2 rows and columns:

```
[[1 2]]
```

```
[5 6]]
```

11. take array of 10 elements and Replace all odd numbers in a NumPy array with -1
12. Find the common items between two arrays
13. Perform arithmetic operations on two NumPy arrays element-wise
 - Add two NumPy arrays element by element.
 - Multiply two NumPy arrays element by element.
14. Write a code to compute the multiplication of two 2x2 NumPy arrays
15. Compute the mean, median, and standard deviation of a NumPy array
 - Mean: Average of numbers.
 - Median: Middle value when sorted.
 - Standard Deviation: How spread out values are from the mean.
16. Remove common items from array

Given:

```
import numpy as np
```

```
a = np.array([1, 2, 3, 4, 5])  
b = np.array([5, 6, 7, 8, 9])
```

17. Create a random 3×2 matrix and find its maximum and minimum values
18. Sorting a NumPy array based on a specific column.

```
import numpy as np  
sampleArray = np.array([[34, 43, 73], [82, 22, 12], [53, 94, 66]])
```

19. Create a random 3×2 matrix and Delete the second column from a given array and insert the following new column in its place.
20. Create an 2D array of 4 x 4 and Swap column 1 and 2 in a 2D array
21. take an array and Sort the rows of a 2D array based on the values of the second column
22. Write a code to Convert a 2D array into a 1D array.
23. Split a 1D array of 9 elements into 3 equal parts.
24. Perform Addition and Squaring on Arrays
25. Count the number of occurrences of each unique element

Given:

```
import numpy as np  
arr = np.array([2, 3, 2, 5, 3, 3, 2, 5])
```

26. Take a matrix of 3 x 3 and Transposing a matrix
27. take a matrix and find `mean()`, `median()` and standard deviation of an 1D array of 10 different heights of person.

ASSIGNMENT-2(A)

Aim: Pandas -Series and Data element

Series:

1. Write a Pandas program to create and display a one-dimensional array-like object containing an array of data using Pandas module.

2. Write a Pandas program to convert a Panda module Series to Python list and it's type. [Hint : Use of `tolist()` method]

3. Write a Pandas program to add, subtract, multiple and divide two Pandas Series.

Sample Series: [2, 4, 6, 8, 10], [1, 3, 5, 7, 9]

Hint: `ds = ds1 + ds2`

`ds = ds1 - ds2`

`ds = ds1 * ds2`

`ds = ds1 / ds2`

4. Write a Pandas program to sort a given Series.

Sample Output:

Original Data Series:

```

0    100
1    200
2  python
3   300.12
4    400

dtype: object

0    100
1    200
3   300.12
4    400

2  python

dtype: object

```

Dataframe:

10. Write a Pandas program to select the rows the score is between 15 and 20 (inclusive).

Sample Python dictionary data and list labels:

```

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James',
'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

```

Expected Output:

Rows where score between 15 and 20 (inclusive):

	attempts	name	qualify	score
c	2	Katherine	yes	16.5
f	3	Michael	yes	20.0
j	1	Jonas	yes	19.0

11. Write a Pandas program to select the rows where number of attempts in the examination is less than 2 and score greater than 15.

Sample Python dictionary data and list labels:

```
exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James',  
'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],  
'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],  
'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],  
'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}  
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

Expected Output:

Number of attempts in the examination is less than 2 and score greater than 15 :

	name	score	attempts	qualify
j	Jonas	19.0	1	yes

12. Write a Pandas program to change the score in row 'd' to 11.5.

Sample Python dictionary data and list labels:

```
exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James',
```



```

'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

```

Expected Output:

Change the score in row 'd' to 11.5:

	attempts	name	qualify	score
a	1	Anastasia	yes	12.5
b	3	Dima	no	9.0
c	2	Katherine	yes	16.5
...				
i	2	Kevin	no	8.0
j	1	Jonas	yes	19.0

13. Write a Pandas program to calculate the sum of the examination attempts by the students.

Sample Python dictionary data and list labels:

```

exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James',
'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

```

Expected Output:

Sum of the examination attempts by the students:

19

14. Write a Pandas program to calculate the mean of all students' scores. Data is stored in a dataframe.

Sample Python dictionary data and list labels:

```
exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James',  
'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],  
'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],  
'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],  
'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}  
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

Expected Output:

Mean score for each different student in data frame:

13.5625

15. Write a Pandas program to append a new row 'k' to data frame with given values for each column. Now delete the new row and return the original DataFrame.

Sample Python dictionary data and list labels:

```
exam_data = {'name': ['Anastasia', 'Dima', 'Katherine', 'James',  
'Emily', 'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],  
'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
```

'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}]}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

Values for each column will be:

name : "Suresh", score: 15.5, attempts: 1, qualify: "yes", label: "k"

Expected Output:

Append a new row:

Print all records after insert a new record:

	attempts	name	qualify	score
a	1	Anastasia	yes	12.5
b	3	Dima	no	9.0
.....				
j	1	Jonas	yes	19.0
k	1	Suresh	yes	15.5

Delete the new row and display the original rows:

	attempts	name	qualify	score
a	1	Anastasia	yes	12.5
b	3	Dima	no	9.0
.....				
i	2	Kevin	no	8.0
j	1	Jonas	yes	19.0

16. Write a Pandas program to get the items of a given series not present in another given series.

Sample Output:

Original Series:

sr1:

0 1

1 2

2 3

3 4

4 5

dtype: int64

sr2:

0 2

1 4

2 6

3 8

4 10

dtype: int64

Items of sr1 not present in sr2:

0 1

2 3

4 5

```
dtype: int64
```

17. Write a Pandas program to get the items which are not common of two given series.

Sample Output:

Original Series:

sr1:

```
0    1
```

```
1    2
```

```
2    3
```

```
3    4
```

```
4    5
```

```
dtype: int64
```

sr2:

```
0    2
```

```
1    4
```

```
2    6
```

```
3    8
```

```
4   10
```

```
dtype: int64
```

ASSIGNMENT-2(B)

Aim:Pandas Library using CSV file

1. Write a python code to show the first 5 rows of the dataframe. (Hint: Use a Head() function)
2. Write a python code to show a last 5 rows of the dataframe (Hint : Use a Tail() function)
3. Write a python code to show the total no. of rows and no. of column of the data frame (Hint: Use a shape method)
4. Write a python code to find the index of the dataframe. (Hint: Use a Index method)
5. Write a python code to display only the column name. (Hint: Use Column method)
6. Write a python code to show the data-type of each column. (Hint: Use dtype method)
7. Write a python code to show basic information about the dataframe. (Hint: Use a info() function)
8. Write a python code to show a total no. of values in each column. (Hint: Use a count() function)
9. Write a python code to show all the unique values of each column. (Hint: Use dataframe['column_name'].unique())
10. Find all the unique “wind speed” values in the data. (Hint : Use dataframe['Column_Name'].nunique())
11. Find the number of times when the weather is exactly clear. (Hint: Use dataframe.groupby('Column_name').get_group('column_name'))
12. Find a number of times when the “wind speed” was exactly 4 km/h. (Hint: dataframe[dataframe['Column_name'] == 4])
13. Find out all the null values in the data. (Hint: Use dataframe.isnull().sum())
14. Find all the instances of when “wind speed is above 24” and “visibility is 25” (Hint: dataframe[(dataframe['column_name'] > 24) & (df['column_name'] == 25)])
15. Find the mean value of each column against each “weather condition”. (Hint: dataframe.groupby('column_name').mean())

16. Find the min and max value of each column against each “weather condition”(Hint: df.groupby('Weather').max())